

Reflection theory for coquasi-Hopf algebras

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The goal of this talk

- The goal of this talk is to introduce the **reflection theory for coquasi-Hopf algebras**. I will explain why this theory is important to the classification of finite-dimensional Nichols algebras and pointed coquasi-Hopf algebras.
- In this report, (H, Φ) is always a coquasi-Hopf algebra with bijective antipode, ${}^H_H\mathcal{YD}$ is the left-left Yetter-Drinfeld module category. $\text{char}(\mathbb{k}) = 0$, $\mathbb{k} = \overline{\mathbb{k}}$.
- The main result: Given a tuple $(\theta \geq 2)$ of finite-dimensional simple Yetter-Drinfeld modules

$$(M_1, M_2, \dots, M_\theta) \in {}^H_H\mathcal{YD} \text{ admitting all reflections} \implies \text{Cartan graph } \mathcal{G}(M).$$

- As an application, we obtain some new criteria for $\dim \mathcal{B}(M_1 \oplus M_2 \oplus \dots \oplus M_\theta) < \infty$.
- Motivation: An important step in lifting method : Given a finite group G , how to classify finite-dimensional Nichols algebra in ${}^G_G\mathcal{YD}$? Heckenberger characterized the missing information via introducing semi-Cartan graph and Weyl groupoid.

Semi-Cartan graph and Weyl groupoid

- Let θ be a natural number, $\mathbb{I} = \{1, \dots, \theta\}$, $\mathcal{X}$ a non-empty set, $(r_i)_{i \in \mathbb{I}}$ a family of maps $r_i : \mathcal{X} \rightarrow \mathcal{X}$, and $(A^X)_{X \in \mathcal{X}}$ a family of $\theta \times \theta$ generalized Cartan matrices.

Definition (Semi-Cartan graph)

The quadruple $\mathcal{G} = \mathcal{G}(\mathbb{I}, \mathcal{X}, (r_i)_{i \in \mathbb{I}}, (A^X)_{X \in \mathcal{X}})$ is called a **semi-Cartan graph** if the following axioms hold.

(CG1) For all $i \in \mathbb{I}$, the map r_i satisfies $r_i^2 = \text{id}_{\mathcal{X}}$.

(CG2) For all $i \in \mathbb{I}$, $X \in \mathcal{X}$, A^X and $A^{r_i(X)}$ have the same i -th row, that is $a_{ij}^X = a_{ij}^{r_i(X)}$ for all $j \in \mathbb{I}$.

- Reflection map: Let $\{\alpha_i\}$ be standard basis of $\mathbb{Z}^\theta$ and $s_i^X \in \text{Aut}(\mathbb{Z}^\theta)$ be the reflection map defined by

$$s_i^X(\alpha_j) = \alpha_j - a_{ij}^X \alpha_i.$$

- Weyl groupoid $\mathcal{W}(\mathcal{G})$** : The groupoid with objects X and morphisms generated by $s_i^X : X \rightarrow r_i(X) \in \text{Hom}(X, r_i(X))$.

Semi-Cartan graph and Weyl groupoid

- The set of real roots: $\Delta^{X, \text{re}} = \{\omega(\alpha_i) \in \mathbb{Z}^{\mathbb{I}} \mid \omega \in \text{Hom}(\mathcal{W}(\mathcal{G}(M)), X), i \in \mathbb{I}\}$. We call $\mathcal{G}$ finite if $\Delta^{X, \text{re}}$ is finite for each $X \in \mathcal{X}$.
- We call $\mathcal{G}$ **standard**, if $A^X = A^Y$ for all $X, Y \in \mathcal{X}$.
- Let $\mathfrak{g} = \mathfrak{g}(A)$ be a **Kac-Moody Lie algebra** determined by generalized Cartan matrix $A = (a_{ij})_{1 \leq i, j \leq n}$. Now let $\mathbb{I} = \{1, \dots, n\}$, $\mathcal{X} = \{X\}$, $r_i(X) = X$, $a_{ij}^X = a_{ij}$ for all $i, j \in \mathbb{I}$. It gives rise to a semi-Cartan graph $\mathcal{G} = \mathcal{G}(\mathbb{I}, \{X\}, (r_i)_{i \in \mathbb{I}}, A)$ and $\text{Hom}(\mathcal{W}(\mathcal{G}), X) = W(A)$.
- This example implies that the concept of semi-Cartan graph and Weyl groupoid generalize the notion of a Cartan matrix and its Weyl group.

Well-known results from Hopf algebra perspective

- (Invent. 2006) Heckenberger introduced Weyl groupoids to obtain finiteness properties of Nichols algebras of diagonal type. In particular, he proved for a Nichols algebra of Cartan type (always give rise to standard semi-Cartan graph)

$$\dim \mathcal{B}(M_1 \oplus \cdots M_\theta) < \infty \iff \dim \mathcal{B}(M_i) < \infty \text{ for each } i,$$

the associated Cartan matrix A^M is of **finite type**



- (Ann. Math. 2010) Andruskiewitsch-Schneider classified all finite-dimensional pointed Hopf algebras over abelian groups provided the group order is not divided by 2, 3, 5, 7.

Well-known results from Hopf algebra perspective

- Any Nichols algebra of diagonal type (equivalently, over abelian groups) defines an arithmetic root system.
- Heckenberger classified all finite arithmetic root systems. (Adv. Math. 2010).



- This work classified Nichols algebras of diagonal type. Furthermore, Angiono gave PBW basis and presentation by generators and relations of all Nichols algebras of diagonal type with finite root system. (J. Reine Angew. Math. 2013, JEMS 2015) is finite.

Well-known results from Hopf algebra perspective

- A very general definition of the semi-Cartan graph and Weyl groupoid have been obtained by Andruskiewitsch-Heckenberger-Schneider. (Amer. J. Math 2010).
- Let H' be a Hopf algebra with bijective antipode. It was proved that a tuple of finite-dimensional simple Yetter-Drinfeld modules in ${}^{H'}_H\mathcal{YD}$ admitting all reflections gives rise to a semi-Cartan graph. (Amer. J. Math 2010, Adv. Math. 2013)
- Cuntz-Heckenberger classified all finite semi-Cartan graphs. (J. Reine Angew. Math 2015)



- Heckenberger-Vendramin classified all finite-dimensional Nichols algebras of rank 2 (JEMS 2017) and rank ≥ 3 (JEMS 2017) over finite non-abelian groups.

From Hopf algebras to coquasi-Hopf algebras

- A coquasi-Hopf algebra H is a coalgebra (H, Δ, ε) equipped with a compatible **quasi-algebra structure** and a quasi-antipode (S, α, β) (may not be unique).
- Multiplication map m and unit map μ are coalgebra homomorphisms, but m is always **nonassociative**!
- A convolution-invertible map $\Phi : H^{\otimes 3} \rightarrow \mathbb{k}$ called **associator**, satisfying

$$\begin{aligned}a_1(b_1c_1)\Phi(a_2, b_2, c_2) &= \Phi(a_1, b_1, c_1)(a_2b_2)c_2, \\ \Phi(a_1, b_1, c_1d_1)\Phi(a_2b_2, c_2, d_2) &= \Phi(b_1, c_1, d_1)\Phi(a_1, b_2c_2, d_2)\Phi(a_2, b_3, c_3), \\ &\dots\end{aligned}$$

for $a, b, c, d \in H$.

- $S : H \rightarrow H$ is a coalgebra antimorphism. We say H has a bijective antipode if S is bijective. $\alpha, \beta : H \rightarrow \mathbb{k}$ are only functions.
- All Hopf algebras are coquasi-Hopf algebras.
- **Comod**(H) is a tensor category.

On finite-dimensional coquasi(quasi)-Hopf algebras

- Drinfeld defined quasi-Hopf algebras in 1989.
- The twisted quantum double $D^\Phi(G)$ was introduced by Dijkgraaf-Pasquier-Roche as the first non-trivial quasi-Hopf algebra, which has a deep relation with CFT. Recent work by Dong-Ng-Ren (Adv.Math.2025) proved that there always exists a holomorphic VOA $\mathbb{V}$ such that $\text{Rep}(\mathbb{V}^G) \cong \text{Rep}(D^\Phi(G))$ as modular tensor categories.
- Etingof-Gelaki (JPAA.2006) and Angiono (Adv.Math.2010) classified all finite-dimensional basic quasi-Hopf algebras over cyclic groups.
- Huang-Liu-Yang-Ye (J. Reine Angew. Math.2020, TAMS.2021, Adv.Math.2026) obtained a complete classification of finite-dimensional coradically graded pointed coquasi-Hopf algebras over abelian groups.
- The theory of semi-Cartan graph for coquasi-Hopf algebras **has not been established before**. Hence, in our settings, we can not deal with Nichols algebras of non-diagonal type before.

Yetter-Drinfeld modules

- A left-left Yetter-Drinfeld module over H is a comodule (V, δ_V) , equipped with a $\mathbb{k}$ -linear map (call it action for simplicity) $\triangleright : H \otimes V \rightarrow V$, satisfying for all $h, l \in H$ and $v \in V$:

$$(hl) \triangleright v = \frac{\Phi(h_2, (l_2 \triangleright v_0)_{-1}, l_3)}{\Phi(h_1, l_1, v_{-1})\Phi((h_3 \triangleright (l_2 \triangleright v_0)_0)_{-1}, h_4, l_4)} (h_3 \triangleright (l_2 \triangleright v_0)_0)_0, \quad (1)$$

$$1_H \triangleright v = v, \quad (2)$$

$$(h_1 \triangleright v)_{-1} h_2 \otimes (h_1 \triangleright v)_0 = h_1 v_{-1} \otimes (h_2 \triangleright v_0). \quad (3)$$

- The category ${}^H_H \mathcal{YD} \cong \mathcal{Z}_r({}^H \mathcal{M})$ as [braided monoidal categories](#), where $\mathcal{Z}_r({}^H \mathcal{M})$ is the right center of the category of left H -comodules.
- Equation (1) follows from hexagon axiom of $\mathcal{Z}_r({}^H \mathcal{M})$.
- The braiding $c_{V,W} : V \otimes W \rightarrow W \otimes V$ is still given by

$$c(v \otimes w) = v_{-1} \triangleright w \otimes v_0.$$

Bosonization and Radford projection

- If R is a Hopf algebra in ${}^H_H\mathcal{YD}$, then $R\#H$ is a coquasi-Hopf algebra.
- More complicated operations:

$$(r\#h)(s\#k) = \frac{\Phi(h_2, s_{-1}, k_2)\Phi(r_{-1}, (h_3 \triangleright s_0)_{-1}, h_5 k_4)}{\Phi(r_{-2}, h_1, s_{-2} k_1)\Phi((h_3 \triangleright s_0)_{-2}, h_4, k_3)} r_0 (h_3 \triangleright s_0)_0 \# h_6 k_5,$$

$$\Delta(r\#h) = \Phi^{-1}(r_{-1}^1, r_{-2}^2, h_1) r_0^1 \# r_{-1}^2 h_2 \otimes r_0^2 \# h_3.$$

- If M is a coquasi-Hopf algebra with surjection $\pi : M \rightarrow H$ and injection $\iota : H \rightarrow M$ such that $\pi \circ \iota = \text{id}_H$, then $R := M^{\text{co}H}$ is a Hopf algebra in ${}^H_H\mathcal{YD}$.
- The Yetter-Drinfeld module structure of R :

$$\text{ad}(h)(r) := \Phi_H(h_1 r_{-1}, S(h_3)_1, h_4)(h_2 r_0) S(h_3)_2,$$

$$\delta_R(r) := \pi(r_1) \otimes r_2.$$

Here $S = \beta * S * \alpha$.

Nichols algebras

- For a given object $V \in {}^H_H\mathcal{YD}$, there are 3 equivalent definitions of Nichols algebra $\mathcal{B}(V)$.
- (1) (i) $\mathcal{B}(V) = \bigoplus_{n \geq 0} \mathcal{B}(V)(n)$ as $\mathbb{N}_0$ -graded Hopf algebras in ${}^H_H\mathcal{YD}$.
(ii) $\mathcal{B}(V)(0) = \mathbb{k}$, $\mathcal{B}(V)(1) = V$.
(iii) $\mathcal{B}(V)$ is generated by V as an algebra in ${}^H_H\mathcal{YD}$.
(iv) $P(\mathcal{B}(V)) = V$.
- (2) Let $I(V) \in {}^H_H\mathcal{YD}$ be the largest Hopf ideal of $T(V)$ contained in $\bigoplus_{n \geq 2} T^n(V)$. The Nichols algebra of V is defined to be

$$\mathcal{B}(V) := T(V)/I(V).$$

- (3) The Nichols algebra $\mathcal{B}(V)$ can be defined by braided symmetrizer

$$\mathcal{B}(V) = \bigoplus_{n \geq 0} T^n(V) / \ker(\mathfrak{S}_n).$$

Here $\mathfrak{S}_n \in \text{End}(V^{\otimes n})$.

Reflection of Yetter-Drinfeld modules

Difficulties: Since ${}^H_H\mathcal{YD}$ is non-strict, many well-known results in Hopf algebra case need to be proved in our setting via different approaches.

Construct reflections:

- Let $A(M_i) := \mathcal{B}(M_i) \# H$ be the bosonization coquasi-Hopf algebra, and $K_i := \mathcal{B}(M)^{\text{co } \mathcal{B}(M_i)}$ a Hopf algebra in ${}^{A(M_i)}_{A(M_i)}\mathcal{YD}$ with 'action' $\text{ad} : A(M_i) \otimes K_i \rightarrow K_i$. Furthermore, $M_j \subseteq K_i$ for each $j \neq i$.
- $L_j := \text{ad}(A(M_i))(M_j) = \text{ad}(\mathcal{B}(M_i))(M_j) = \bigoplus_{n \geq 0} \text{ad}(M_i)^n(M_j) \in {}^{A(M_i)}_{A(M_i)}\mathcal{YD}$.

Reflection of Yetter-Drinfeld modules

Construct reflections:

- We say the tuple M **admits the i -th reflection** if for all $j \neq i$ there is a natural number $m_{ij}^M \geq 0$ such that $(\operatorname{ad} M_i)^{m_{ij}^M}(M_j) \neq 0$ and $(\operatorname{ad} M_i)^{m_{ij}^M+1}(M_j) = 0$.
- If M admits the i -th reflection, we set $R_i(M) = (\tilde{M}_1, \dots, \tilde{M}_\theta)$, where

$$\tilde{M}_j = \begin{cases} M_i^*, & \text{if } j = i, \\ \operatorname{ad}(M_i)^{m_{ij}^M}(M_j), & \text{if } j \neq i. \end{cases}$$

Each $\tilde{M}_j$ is proved to be simple in ${}^H_H\mathcal{YD}$ for $1 \leq j \leq \theta$.

- We define $a_{ii}^M = 2$ for all $1 \leq i \leq \theta$ and define $a_{ij}^M = -m_{ij}^M$. Then $(a_{ij}^M)_{i,j \in \mathbb{I}}$ is proved to be a generalized Cartan matrix.
- We say $M \cong N$ in if $M_i \cong N_i$ for each $i \in \mathbb{I}$. If M admits the i -th reflection for some $i \in \mathbb{I}$, so does N . If $M \cong N$, $R_i(M) \cong R_i(N)$ and $a_{ij}^M = a_{ij}^N$ for each $j \in \mathbb{I}$. The isomorphism class of M is denoted by $[M]$.

Semi-Cartan graph structure

- We say M admits all reflection sequences if M admits reflection sequence $(i_1, i_2, \dots, i_l)$ for all $l \in \mathbb{N}_0$ and $i_1, i_2, \dots, i_l \in \mathbb{I}$. In this case, we denote

$$\mathcal{F}_\theta(M) = \{R_{i_1}(\dots(R_{i_l}(M))\dots) \mid l \in \mathbb{N}_0, i_1, \dots, i_l \in \mathbb{I}\}.$$

Theorem

Suppose M admits all reflections. We define the set $\mathcal{X} = \{[P] \mid P \in \mathcal{F}_\theta(M)\}$, and the map

$$r : \mathbb{I} \times \mathcal{X} \rightarrow \mathcal{X}, (i, [X]) \mapsto [R_i(X)].$$

Then

$$\mathcal{G}(M) = (\mathbb{I}, \mathcal{X}, r, (A^{[X]})_{[X] \in \mathcal{X}}),$$

where $A^{[X]} = (a_{ij}^{[X]})_{i,j \in \mathbb{I}}$ for all $[X] \in \mathcal{X}$, is **a semi-Cartan graph**.

Reflections of Nichols algebras

- The above result does not involve Nichols algebras!
- Actually, it is a byproduct of the following isomorphism of Hopf algebras.

Theorem

Suppose M admits the i -th reflection, then there is an isomorphism of Hopf algebras in ${}^H_H\mathcal{YD}$:

$$\mathcal{B}(R_i(M)) \cong \Omega_i \left(\mathcal{B}(M)^{\text{co } \mathcal{B}(M_i)} \right) \# \mathcal{B}(M_i^*).$$

- Here Ω_i was proved to be a **braided monoidal equivalence**:

$$\begin{aligned} \Omega_i : {}_{\mathcal{B}(M_i)}^{\mathcal{B}(M_i)}\mathcal{YD}(C)_{\text{rat}} &\longrightarrow {}_{\mathcal{B}(M_i^*)}^{\mathcal{B}(M_i^*)}\mathcal{YD}(C)_{\text{rat}}. \\ (V, \delta, \lambda) &\mapsto (V, \delta', \lambda'). \end{aligned}$$

Here $C = {}^H_H\mathcal{YD}$.

Some remarks on Ω_i

- An object $V \in {}^{\mathcal{B}(M_i)}_{\mathcal{B}(M_i)}\mathcal{YD}(C)$ is called rational if for any $v \in V$, there exists $n_0 \in \mathbb{N}_0$, such that $\mathcal{B}(M_i)(n) \triangleright v = 0$ for $n > n_0$.
- The category ${}^{\mathcal{B}(M_i)}_{\mathcal{B}(M_i)}\mathcal{YD}(C)_{\text{rat}}$ is a **braided monoidal subcategory** of ${}^{\mathcal{B}(M_i)}_{\mathcal{B}(M_i)}\mathcal{YD}(C)$.
- Why must rational Yetter-Drinfeld module: ${}^{\mathcal{B}(M_i)}_{\mathcal{B}(M_i)}\mathcal{YD}(C) \not\cong {}^{\mathcal{B}(M_i^*)}_{\mathcal{B}(M_i^*)}\mathcal{YD}(C)$
- A generalization: comodules $\leftrightarrow$ rational modules.

Some remarks on Ω_i

- The braided monoidal equivalence is determined by the following **non-degenerate Hopf pairing**

$$\begin{aligned}\omega_i &: \mathcal{B}(M_i^*) \otimes \mathcal{B}(M_i) \rightarrow \mathbb{k}, \\ \omega_i \mid_{M_i^{\otimes n}} &= \text{ev}_{M_i^{\otimes n}} \circ (\text{id} \otimes \mathfrak{S}_n).\end{aligned}$$

More generally, if A, B are locally finite $\mathbb{N}_0$ -graded Hopf algebras in ${}^H_H\mathcal{YD}$ with bijective antipode, and there is a non-degenerate Hopf pairing $A \otimes B \rightarrow \mathbb{k}$. Then there is a braided monoidal equivalence:

$$\Omega : {}^B_B\mathcal{YD}(C)_{\text{rat}} \rightarrow {}^A_A\mathcal{YD}(C)_{\text{rat}}.$$

- The condition M admits i -th reflection is equivalent to $\mathcal{B}(M)^{\text{co } \mathcal{B}(M_i)} \in {}^{\mathcal{B}(M_i)}_{\mathcal{B}(M_i)}\mathcal{YD}(C)_{\text{rat}}$.

A sketch proof

- There is a braided monoidal isomorphism: $\frac{\mathcal{B}(M_i)}{\mathcal{B}(M_i)} \mathcal{YD}(C) \cong \frac{A(M_i)}{A(M_i)} \mathcal{YD}.$
- Let $(V, \triangleright, \delta)$ be an object in $\frac{A(M_i)}{A(M_i)} \mathcal{YD}.$ We define:

$$\mathcal{TV} = \{v \in V \mid A(M_i)(n) \triangleright v = 0 \text{ for all } n > 0\},$$

$$\mathcal{DV} = \{v \in V \mid \delta(v) \in H \otimes V\}.$$

- We have $\mathcal{T}\Omega_i(V) = \mathcal{DV}$ and $\mathcal{D}\Omega_i(V) = \mathcal{T}(V).$
- The condition M admits i -th reflection is equivalent to $\mathcal{B}(M)^{\text{co } \mathcal{B}(M_i)} \in \frac{A(M_i)}{A(M_i)} \mathcal{YD}_{\text{rat}}.$

A sketch proof

- Let $N = \bigoplus_{j \neq i} M_j$. We proved the following properties.
 - (1) $\mathcal{B}(M)^{\text{co } \mathcal{B}(M_i)} \cong \mathcal{B}(\text{ad}(\mathcal{B}(M_i))(N)) \cong \mathcal{B}(\bigoplus_{j \neq i} L_j)$ as Nichols algebras.
 - (2) $\Omega_i(\mathcal{B}(\bigoplus_{j \neq i} L_j)) \cong \mathcal{B}(\Omega_i(\bigoplus_{j \neq i} L_j))$ as Nichols algebras.
 - (3) $\mathcal{B}(\Omega_i(\bigoplus_{j \neq i} L_j)) \# \mathcal{B}(M_i^*) \cong \mathcal{B}(\mathcal{D}\Omega_i(\bigoplus_{j \neq i} L_j) \oplus M_i^*)$ as Nichols algebras.
 - (4) $\mathcal{T} \bigoplus_{j \neq i} L_j = \bigoplus_{j \neq i} \tilde{M}_j$.
- Combining (1)-(4) together, we proved

$$\mathcal{B}(R_i(M)) \cong \Omega_i \left(\mathcal{B}(M)^{\text{co } \mathcal{B}(M_i)} \right) \# \mathcal{B}(M_i^*).$$

A sketch proof

- In the proof of (3), we have

$$\Omega_i\left(\bigoplus_{n=0}^{m_{ij}^M} \text{ad}(M_i)^n(M_j)\right) = \Omega_i(\text{ad}(\mathcal{B}(M_i)(M_j))) \cong \text{ad}(\mathcal{B}(M_i^*)(\tilde{M}_j)) = \bigoplus_{n=0}^{m_{ij}^{R_i(M)}} \text{ad}(M_i^*)^n(\tilde{M}_j).$$

This implies $m_{ij}^M = m_{ij}^{R_i(M)}$ and proves (CG2).

- We have

$$\begin{aligned} M_j &= \mathcal{D} \text{ad}(\mathcal{B}(M_i)(M_j)) = \mathcal{T} \Omega_i(\text{ad}(\mathcal{B}(M_i)(M_j))) \\ &\cong \mathcal{T} \text{ad}(\mathcal{B}(M_i^*)(\tilde{M}_j)) = \text{ad}(M_i^*)^{m_{ij}^{R_i(M)}}(\tilde{M}_j) = \widetilde{\tilde{M}_j}. \end{aligned}$$

Note that $(M_i^*)^* \cong M_i$, which proves (CG1).

- Therefore, if M **admits all reflections**, then M gives rise to a semi-Cartan graph.

From semi-Cartan graphs to Cartan graphs

- We introduce an equivalent definition given by Heckenberger-Schneider.
- Let $X \in \mathcal{X}$, $l \geq 0$, and $\kappa = (i_1, \dots, i_l) \in \mathbb{I}^l$. Let

$$\Lambda^X(\kappa) = \{\text{id}_X s_{i_1} \cdots s_{i_{k-1}}(\alpha_{i_k}) \mid 1 \leq k \leq l\}.$$

- We say that κ is **X-reduced** if for any $1 \leq k < l$, $\alpha_{i_k} \notin \Lambda^{r_{i_k} \cdots r_{i_1}(X)}(i_{k+1}, \dots, i_l)$.
- (CG3'): For any $X \in \mathcal{X}$ and any X-reduced sequence κ , $\Lambda^X(\kappa) \subseteq \mathbb{N}_0^{\mathbb{I}}$.
- Now we consider the longest reduced sequence of the form $\kappa_{ij} = (i, j, i, j, \dots)$. We use $n_{ij} := \text{length of } \kappa_{ij}$.
- (CG4'): For any $X \in \mathcal{X}$ and any $i, j \in \mathbb{I}$ with $i \neq j$ and $n_{ij}^X < \infty$, we have

$$(r_i r_j)^{n_{ij}^X}(X) = X, \quad \text{id}_X(s_i s_j)^{n_{ij}^X}(\alpha_k) = \alpha_k$$

for all $k \neq i, j$.

- We say a semi-Cartan graph $\mathcal{G}$ a Cartan graph if $\mathcal{G}$ satisfies (CG3')+(CG4').

From semi-Cartan graphs to Cartan graphs

- Why we need Cartan graphs?
In my opinion, it is necessary when we consider **PBW basis** of Nichols algebras.
- Finite Cartan graph $\implies$ Existence of unique longest elements in Weyl groupoid $\implies$ convex order of real roots $\implies$ PBW basis.
- Key tools: An analogue of "Lusztig isomorphism":

$$T_i^M = \mathcal{S}_{K_i^M}^{-1} \circ \mathcal{S}_{\mathcal{B}(R_i(M))} : L_i^{R_i(M)} := {}^{\text{co } \mathcal{B}(M_i^*)} \mathcal{B}(R_i(M)) \rightarrow K_i^M$$

T_i^M **preserves $\mathbb{N}_0^\theta$ -grading** via $\deg(T_i^M(L_i^{R_i(M)}(\alpha))) = s_i^M(\alpha)$

- This isomorphism is important when determining generating relations of Nichols algebras of diagonal type (over Hopf algebras).

A sketch proof

- (CG3'): Given a reduced sequence $\kappa = (i_1, \dots, i_l)$. By applying the above isomorphism repeatedly, we can construct an irreducible object $M_{\beta_k} \hookrightarrow \mathcal{B}(M)$ with degree $\beta_k = s_{i_1} s_{i_2} \cdots s_{i_{k-1}}(\alpha_{i_k})$ from $R_{(i_1, \dots, i_{k-1})}(M)_{i_k}$ for $1 \leq k \leq l$. Then $\beta_k \in \mathbb{N}_0^\theta$ since T_i^M preserves $\mathbb{N}_0^\theta$ -grading.
- (CG4'): Consider the following two reduced sequence $\kappa_{ij} = (i, j, i, j, \dots)$ and $\kappa_{ji} = (j, i, j, i, \dots)$. Then we can define a well-defined map

$$\phi_k : R_{\kappa_{ij}}(M)_k \longrightarrow M \longrightarrow R_{\kappa_{ji}}(M)_k$$

for each $k \neq i, j$ by using T_i^M repeatedly, which corresponds to $(s_i s_j)^{n_{ij}}$. We then show ϕ_k is an isomorphism of objects and fix $\mathbb{N}_0^\theta$ -grading via some combinatorial arguments. Therefore we prove (CG4').

Some criteria

- If $\dim \mathcal{B}(M) < \infty$, then M admits all reflections and $\mathcal{G}(M)$ must be finite. Furthermore, $\dim \mathcal{B}(R_i(M)) = \dim \mathcal{B}(M)$ for each $i \in \mathbb{I}$.
- If M admits i -th reflection, then $\text{GKdim } \mathcal{B}(M) = \text{GKdim } \mathcal{B}(R_i(M))$.
- If $\mathcal{G}(M)$ is standard, then $\text{Hom}(\mathcal{W}(\mathcal{G}(M)), X)$ coincide with Weyl group $W(A^M)$ of A^M . We proved $\dim \mathcal{B}(M) < \infty$ is equivalent to each $\mathcal{B}(M_j)$ being finite-dimensional and A^M being a **Cartan matrix of finite type**.
- The following are equivalent.
 - (1) $\mathcal{B}(M)$ is **finite-dimensional**.
 - (2) M admits all reflections, $\mathcal{G}(M)$ is finite and $\mathcal{B}(P_i)$ is finite-dimensional for all $P \in \mathcal{F}_\theta(M)$ and $i \in \mathbb{I}$.

Application I: A non-diagonal type example

- On classification of finite-dimensional Nichols algebras in ${}^G_G\mathcal{YD}^\Phi$, where G abelian, Φ 3-cocycle on G .
- **Non-diagonal type Nichols algebras** do occur! We focus on the following class. It was proved to be infinite-dimensional through heavy computations by Huang-Liu-Yang-Ye (Adv. Math. 2026). We aim to provide an alternative proof.
- Let $G = \mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2 = \langle g_1 \rangle \times \langle g_2 \rangle \times \langle g_3 \rangle$, and

$$\Phi \left(g_1^{i_1} g_2^{i_2} g_3^{i_3}, g_1^{j_1} g_2^{j_2} g_3^{j_3}, g_1^{k_1} g_2^{k_2} g_3^{k_3} \right) = (-1)^{k_1 j_2 i_3},$$

be a 3-cocycle on G , where $0 \leq i_l, j_l, k_l \leq 1, 1 \leq l \leq 3$.

- In the category ${}^G_G\mathcal{YD}^\Phi$, there are six pairwise non-isomorphic Yetter-Drinfeld modules W_i , such that $\mathcal{B}(W_i)$ is finite-dimensional, $1 \leq i \leq 6$.

Application I: A non-diagonal type example

- Let (W_i, W_j) be any 2-tuple, $1 \leq i < j \leq 6$.

Case (i): $\deg(W_i) \cdot \deg(W_j) = g_1 g_2 g_3$, then $\text{ad}(W_i)^n(W_j) \neq 0$ for all $n \in \mathbb{N}_0$. Thus $\mathcal{B}(M_i \oplus M_j)$ is infinite-dimensional.

Case (ii): $\deg(W_i) \cdot \deg(W_j) \neq g_1 g_2 g_3$, then one can show that (W_i, W_j) is a standard Cartan graph with Cartan matrix A_2 and $\mathcal{B}(M_i \oplus M_j)$ is of dimension 64.

- Let $W = (W_i, W_j, W_k)$ be any 3-tuple, $1 \leq i < j < k \leq 6$.

We only need to consider the case $\deg(W_i) \cdot \deg(W_j) \cdot \deg(W_k) = g_1 g_2 g_3$.

Then W admits all reflections and $\mathcal{G}(W)$ is a **standard Cartan graph**. The generalized Cartan matrix A^W is

$$A_2^{(1)} = \begin{pmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{pmatrix}$$

Therefore the Nichols algebra $\mathcal{B}(W)$ is infinite-dimensional.

Application I: A non-diagonal type example

- For each $X \in \mathcal{X}$, the set $\Delta^{X, \text{re}}$ coincides with the set of real roots of Lie algebra $\mathfrak{g}(A_2^{(1)})$, which is infinite.
- We proved $\mathcal{B}(W)$ is an **affine Nichols algebra**. That is, the Tits cone of $\mathcal{G}(W)$ under crystallographic Tits arrangements is a half plane.
- This means we realize an affine Nichols algebra over a coquasi-Hopf algebra, which broadens the approaches to realize affine Nichols algebras.

Application II: Diagonal type

- The second application aims to show that the set of Cartan graphs given by Nichols algebras over coquasi-Hopf algebras of diagonal type coincides with Hopf case.
- If $M \in {}^H_H\mathcal{YD}$ be a Yetter-Drinfeld module of diagonal type. Then we can prove $\text{Supp } M$ generates a finite abelian group G , and $\Phi|_G$ is an abelian (trivializable) 3-cocycle on G .
- Let $M = (M_1, \dots, M_\theta)$ be a θ -tuple of diagonal type. We have

$$\begin{aligned} \Delta(\text{ad}(x_i)^n(x_j)) &= \text{ad}(x_i)^n(x_j) \otimes 1 + 1 \otimes \text{ad}(x_i)^n(x_j) \\ &+ \sum_{k=1}^n \binom{n}{k} \prod_{q_{ii} \mid l=n-k}^{n-1} (1 - q_{ii}^l q_{ij} q_{ji}) \prod_{q=1}^k \Phi(g_i^{q-1}, g_i, g_i^{n-q} g_j) \overrightarrow{x_i}^k \otimes \text{ad}(x_i)^{n-k}(x_j). \end{aligned}$$

- Hence M admits i -th reflection if and only if there is $m \in \mathbb{N}_0$ such that

$$(m+1)_{q_{ii}}(q_{ii}^m q_{ij} q_{ji} - 1) = 0.$$

Application II: Diagonal type

- Let $i \in \mathbb{I}$ and M admits i -th reflection. Assume the braiding matrix of $R_i(M)$ is $(q'_{jk})_{j,k \in \mathbb{I}}$.
- q'_{jk} may involve associators, for example, if $j, k \neq i$, then

$$q'_{jk} = \prod_{l=0}^{-a_{ik}^M - 1} \widetilde{\Phi}_{g_i}^{-a_{ij}^M} (g_i, g_i^l g_k) \prod_{m=0}^{-a_{ij}^M - 1} (\widetilde{\Phi}_{g_i}^{-1}(g_j g_i^m, g_i) q_{ii}^{-a_{ij}^M} q_{ji})^{-a_{ik}^M} \widetilde{\Phi}_{g_k}^{-1}(g_j g_i^m, g_i) q_{ik}^{-a_{ij}^M} q_{jk}.$$

- But they satisfy the following relations.

$$q'_{jj} = q_{jj} q_{ij}^{-a_{ij}^M} q_{ji}^{-a_{ij}^M} (q_{ii}^M)^2,$$

$$q'_{ij} q'_{ji} = q_{ij}^{-1} q_{ji}^{-1} q_{ii}^{2a_{ij}^M},$$

$$q'_{jk} q'_{jk} = (q_{jk} q_{kj}) (q_{ik} q_{ki})^{-a_{ij}^M} (q_{ij} q_{ji})^{-a_{ik}^M} q_{ii}^{2a_{ij}^M a_{ik}^M}.$$

This means reflection **only relies on braiding**.

Application II: Diagonal type

- We investigate the behavior of Cartan graphs under braided monoidal equivalences.
- Let $\mathcal{G} = \mathcal{G}(I, \mathcal{X}, r, A)$ and $\mathcal{G}' = \mathcal{G}(J, \mathcal{Y}, t, B)$ be Cartan graphs. If $\beta : I \rightarrow J$, $\gamma : \mathcal{X} \rightarrow \mathcal{Y}$ be bijective maps such that for all $i, j \in I$ and $X \in \mathcal{X}$,

$$\gamma(r_i(X)) = t_{\beta(i)}(\gamma(X)), \quad a_{ij}^X = b_{\beta(i)\beta(j)}^{\gamma(X)}.$$

Then we call $(\beta, \gamma) : \mathcal{G} \rightarrow \mathcal{G}'$ an isomorphism of Cartan graphs.

- Let H, H' be two coquasi-Hopf algebras with bijective antipode. We fix a **braided monoidal equivalence**:

$$F : {}^H_H \mathcal{YD} \longrightarrow {}^{H'}_{H'} \mathcal{YD}.$$

Theorem

The tuple M admits all reflections if and only if $F(M)$ does. Under this condition, $\mathcal{G}(M)$ is isomorphic to $\mathcal{G}(F(M))$ as Cartan graphs.

Application II: Diagonal type

- Huang-Liu-Yang-Ye provided a method to classify Nichols algebras in ${}^G\mathcal{YD}^\Phi$ of diagonal type.

$$\begin{array}{ccc}
 \text{Given } \mathcal{B}(M) \in {}^G\mathcal{YD}^{\Phi_{\underline{a}}} & & \mathcal{B}(M')^{J_{\underline{a}}} \cong \mathcal{B}(M) \in {}^G\mathcal{YD}^{\Phi_{\underline{a}}} \\
 \downarrow & & \uparrow \text{ Under some condition} \\
 \mathcal{B}(M) \cong \mathcal{B}(\tilde{M}) \in {}^G\mathcal{YD}^{\pi^*(\Phi_{\underline{a}})} & & \mathcal{B}(M')^{J_{\underline{a}}} \in {}^G\mathcal{YD}^{\partial(J_{\underline{a}})} \\
 \downarrow & & \uparrow \\
 \mathcal{B}(\tilde{M}) \text{ twisted equivalent to } \mathcal{B}(\tilde{M})^{J_{\underline{a}}^{-1}} \in {}^G\mathcal{YD} & & \text{Given Nichols algebra } \mathcal{B}(M') \in {}^G\mathcal{YD}
 \end{array}$$

- We assume $M \in {}^G\mathcal{YD}^{\Phi_{\underline{a}}}$ admits all reflections.
- $\tilde{M}$ has the same braiding matrix with M . Then $\tilde{M}$ admits i -th reflections for all i .
- Although $R_i(M)$ and $R_i(\tilde{M})$ have **different braiding matrix** after reflection, but their **Cartan matrix coincide**. Hence $\tilde{M}$ admits all reflections and $\mathcal{G}(\tilde{M}) \cong \mathcal{G}(M)$.

Application II: Diagonal type

- The second step is a twist, which is a special kind of braided monoidal equivalence in categorical sense.
- Furthermore $\mathcal{G}(M)$ is isomorphic to $\mathcal{G}(\tilde{M}^{J_{\underline{a}}^{-1}})$.
- The following two sets of isomorphism classes of Cartan graphs are equal:

$$\mathcal{P} = \left\{ [\mathcal{G}(M)] \left| \begin{array}{l} M = (M_1, \dots, M_\theta) \in {}_{\mathbb{G}}\mathcal{YD}^\Phi, \theta \geq 2, M \text{ admits all reflections,} \\ G \text{ is an abelian group, } \Phi \text{ is an abelian 3-cocycle} \end{array} \right. \right\},$$

$$\mathcal{Q} = \left\{ [\mathcal{G}(V)] \left| \begin{array}{l} M = (V_1, \dots, V_\tau) \in {}_{\mathbb{G}}\mathcal{YD}, \tau \geq 2, V \text{ admits all reflections} \\ \mathbb{G} \text{ is an abelian group,} \end{array} \right. \right\}.$$

Thank You